



UNIVERSITY OF
CAMBRIDGE

Institute for
Sustainability
Leadership

Official Problem Statement

Cambridge Edge AI
Innovation for
Sustainability Challenge
2026

Hosted by



MICROIP

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ARCULUS SYSTEM

Academic partners: University of Cambridge Institute for Sustainability Leadership (CISL)

About this document

This is the official problem statement and scoring reference for the Challenge, intended for participants, judges, and partner organisations alike. Team size, prizes, and registration logistics will be announced separately by the organising administration and are governed by the official announcement; the technical rules and scoring criteria are governed by this document.

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1. Vision

We are at a turning point as AI moves from centralised compute towards edge collaboration. The central question of this Challenge is:

When every node must keep its own raw data, and compute and energy resources are limited, can a system nonetheless—by exchanging the knowledge it has learned—produce a collective intelligence that surpasses the capability of any single node while upholding sustainability values?

This Challenge invites you to design a **sustainability-oriented, multi-node Edge AI learning scheme**. Multiple edge nodes, each with local observations and the ability to learn on their own, must—**without sharing raw data**—demonstrate genuine collective gain through **knowledge exchange, co-evolution, and continual learning**. The form of a node's self-learning is not restricted and also includes **self-reinforcement learning**: the strategies and knowledge a node acquires through self-reinforcement learning may, provided no raw data leaves the node, be offered for other edge nodes to learn from; this is an important form of knowledge exchange.

We invite you to redefine the boundaries of distributed cognition under real-world constraints. Innovations developed through this Challenge may help shape the resilient, energy-efficient edge ecosystems of tomorrow.

1.1 Positioning of the Challenge

This Challenge is a **first step in broadly gathering creative approaches**. The organisers provide no SDK, platform, hardware, or proprietary dataset; participants submit their solutions using resources they already have. We encourage students to **explore freely**, and from their work we aim to converge on directions worth deepening into future collaboration.

2. Problem Statement

Design a multi-node Edge AI learning scheme. Each node:

- May access only its own local data.
- Must not transmit to other nodes any raw data, any content from which raw data could reasonably be reconstructed, or any intermediate representation strongly suspected of data tunnelling.
- May exchange learned knowledge through any mechanism, but every machine-readable exchange of information between nodes must be disclosed and open to review.
- Is not restricted in its form of self-learning: beyond supervised / self-supervised learning, **self-reinforcement learning is also permitted**; the strategies or knowledge a node obtains through self-reinforcement learning **may be offered for other edge nodes to learn from as a form of knowledge exchange**, provided this complies with the exchange rules in §5 (no raw data leaves the node).

- Should be able to handle at least one of the following challenges: Non-IID data distribution, the emergence of new classes, node failure, continual learning, catastrophic forgetting.
- Should demonstrate explicit awareness of energy, compute, or communication costs.

This Challenge sits within the **Edge AI** domain, covering on-device learning and self-adaptation on edge devices. **The Challenge is primarily focused on, but not limited to computer vision (image) tasks. For example:**

- By default, each node's local data are images, and participants may use any **public image dataset** as experimental material (the organisers do not designate an official dataset).
- Participants may also adopt other types of AI tasks (e.g. audio, sensor sequences, multimodal), provided they **explain in detail the techniques used, the data modality, and the rationale**. Those choosing a non-image modality must still explain how their solution maps onto the scoring axes in §4 and the knowledge-exchange rules in §5 (including the prohibition on data tunnelling).

Model architecture, deployment form, communication mechanism, and the type of knowledge representation are all for participants to design freely. The organisers **do not mandate** any particular SDK, Docker, framework, runner, or official platform.

2.1 Suggested (Non-Mandatory) Application Scenarios

To help participants understand the brief, the organisers offer the following suggested scenarios (**choose any, or define your own**; the choice of scenario does not itself affect scoring, but the quality with which it is addressed is assessed under Axis B).

Image-oriented (suggested directions):

- Collaborative image classification / object detection on low-power chips (e.g. the Jetson series, Coral, Raspberry Pi).
- Distributed visual recognition across multiple robots / drones.
- Defect-inspection imagery on smart-factory production lines, collaborating across lines without raw images leaving each line.

Non-image directions (for reference if adopting another modality; you may also define your own):

- Acoustic / vibration anomaly detection on factory machinery, collaborating across machines without raw signals leaving each machine.
- Personalised recognition from wearable or on-device sensor sequences.
- Keyword / event spotting from audio on edge devices.

Non-image tasks are not penalised for choosing not to adopt one of the image scenarios above; what Axis B assesses is whether the chosen scenario is concrete, reasonable, and integrable.

3. Core Challenges

The purpose of this brief is not to test who can produce the most complex network-engineering solution, but **who can design the most effective and most industrially integrable AI learning mechanism under constrained exchange and constrained resources**. Judges will pay particular attention to:

- Under Non-IID conditions, how can multiple nodes learn effectively?
- When raw data cannot be exchanged, which knowledge representations are most valuable to exchange, and at the lowest communication cost?
- How can a model absorb cross-node knowledge while never seeing others' raw data?
- As new classes, new scenarios, or new tasks appear over time, how can plasticity be retained while forgetting is reduced?
- When some nodes go offline, fail, or are otherwise constrained, how does the system as a whole sustain the benefits of collaboration?
- If a node improves itself through reinforcement learning, how can the resulting strategies / knowledge be conveyed to and absorbed by other nodes without raw experience data leaving the node?
- Has the solution been designed appropriately so that it can later be integrated into industrial settings at low cost, and is the adoption path made clear?

4. Scoring Criteria (25 / 25 / 25 / 25)

Axis	Weight	Scope
A. Sustainability Impact	25%	Energy-efficient design, compute and communication savings, measurable environmental benefit
B. Ease of Integration & Feasibility	25%	Clarity of integration path and dependencies, portability to industrial settings, honest identification of adoption barriers, interface-design awareness
C. Technical Maturity	25%	System stability, scenario-driven demonstration and adaptability, evaluation completeness, collaboration-gain measurement, reproducibility
D. Creativity	25%	Innovation in the knowledge-exchange mechanism, cross-domain imagination, future extensibility

4.1 Axis A: Sustainability Impact (25%)

The core of this axis is comparison. Participants must place their "mutually-learning collaborative approach" side by side with a clearly defined "traditional baseline approach", and present the sustainability difference (saving or efficiency) that collaboration brings. **Reporting only the absolute figures of the collaborative approach, with no comparison baseline, cannot earn a high score on this axis.**

Baseline (choose one; the chosen baseline and the reason for it must be stated explicitly in the research document):

- **Baseline A | Centralised cloud:** uses "all nodes' raw data uploaded to a central cloud, training a single large model" as the traditional comparator. The comparison focuses on the **communication volume and energy the collaborative approach saves** (no raw data is transmitted; inference moves to the edge). Suited to solutions whose headline is "avoiding data upload and reducing cloud compute".

Baseline A is a counterfactual baseline used only for comparison—it may be a pure estimate or an offline simulation using public data; the organisers neither require nor encourage actually centralising any node's private raw data. Baseline A must not serve as the initialisation weights of the submitted system or as any part of the collaboration workflow; model initialisation follows §5.3 in all cases.

- **Baseline B | Local-only:** each node works on its own, exchanging no knowledge with the others (the same `local_only` baseline as in Axis C). Collaboration **adds** communication cost, so the comparison focuses on **efficiency**—`gain_per_joule`, `gain_per_byte`, i.e. how much collaboration gain is obtained per additional unit of energy / communication spent. Suited to solutions whose headline is "a small amount of exchange in return for substantial gain".

The two baselines answer different questions; judges score on **how rigorous a team's own comparison is and how reasonable its assumptions are** (relative assessment within the axis; no cross-team comparison of absolute figures).

Axis A sub-criteria (judges assess evidence quality against these; choosing Baseline A or B does not in itself add or deduct marks):

- **A1 Reasonableness of the comparison baseline:** are the chosen baseline and its necessary assumptions reasonable and honest?
- **A2 Completeness of the cost inventory:** are communication, compute, and energy covered to the extent the claim requires (see Appendix E)?
- **A3 Soundness of the directional conclusion:** does collaboration genuinely save, or does it obtain gain at a reasonable cost?
- **A4 Disclosure of uncertainty and counter-cases:** is the uncertainty of the estimate disclosed, and is the case where collaboration may not be worthwhile discussed?

Dimensions to present in the comparison (present those relevant to the chosen baseline):

- **Communication** (no hardware required; recommended for all teams): total exchanged data volume (bytes) and message frequency for the traditional vs collaborative approach; the collaborative approach's communication-saving ratio, or the learning gain obtained per unit of communication cost.
- **Compute** (no hardware required; recommended for all teams): a comparison of model size, parameter count, and computational complexity (FLOPs / MACs) between the two approaches, and the collaborative approach's compute-saving ratio relative to the baseline.

- **Energy estimate:** energy per inference (J/inference) for both approaches, and a comparison of performance and collaboration gain per unit of energy—a theoretical estimate citing literature baselines is acceptable, with the estimation logic and assumptions explained in the research document; if measured values exist, include them too.
- **Environmental-benefit narrative:** if deployed in the target setting (e.g. factory, healthcare, robotics), the estimated benefit of the collaborative versus traditional approach in carbon footprint, waste, or energy. **If this is used as a basis for the Axis A score, the deployment scale, time horizon, the sources of conversion factors, and the uncertainty must be stated; otherwise it serves only as qualitative support and not as a primary basis for scoring.**

Estimation-first; no measurement hardware assumed. The Challenge assumes most participants have no dedicated measurement hardware, so this axis is scored mainly on **reasonable estimation**; judges will prioritise **the rigour of the comparison design and of the estimation method, assumptions, and literature baselines**, not whether you possess measurement instruments. Actually running on a simple edge device (e.g. Raspberry Pi, Jetson, a phone) to obtain measured energy figures is an encouraged bonus, not a requirement; not running on real hardware is not penalised. **A theoretical estimate can earn a high score and is not automatically marked down for the absence of measurement**; however, where the transparency of assumptions is comparable, measured values or a verifiable execution trace can raise the credibility rating. See Appendix E.

4.2 Axis B: Ease of Integration & Feasibility (25%)

This Challenge is intended to gather solutions and prototypes; this axis assesses the "design awareness and quality of argument" around integration feasibility, not product-grade off-the-shelf readiness. Judges will examine:

- **Clarity of the integration path:** does the solution clearly explain how it would integrate into an existing Edge AI pipeline, what prerequisites are needed, and reasonably estimate the effort required?
- **Restraint in dependencies:** are the required frameworks, libraries, and hardware common and clearly versioned, avoiding closed or obscure dependencies?
- **Industrial applicability:** can the solution map to at least one real industrial setting, with the adoption path explained?
- **Honest identification of barriers:** does it honestly point out the integration barriers, dependency risks, and open problems on the way to industrial deployment?
- **Interface-design awareness:** are the inter-node interaction interfaces / boundaries deliberately designed to be easy to connect to later, and are they documented?

4.3 Axis C: Technical Maturity (25%)

Judges will examine:

- **System stability:** does the solution run stably, and can it handle error conditions?

- **Collaboration-gain measurement** (core sub-criterion): does it provide a concrete value for $\text{Collaboration Gain} = \text{Score}(\text{Collective}) - \text{Score}(\text{local_only})$, or, where this cannot be provided, a reasonable alternative comparison? **ΔG must be accompanied by the "Collaboration-Gain Fairness Disclosure Table" (see Appendix A.1)**, stating the test-set scope, data split, and the training budgets of `local_only` and `collective`, among others; where this is missing, or budgets are misaligned without explanation, ΔG may not be treated as high-confidence evidence.
- **Evaluation completeness**: does it include baseline comparisons, ablations, or control groups? Is the comparison fair (baseline and collaborative version using the same architecture, training budget, and tuning effort)?
- **Reproducibility**: from the submitted code and run instructions, can judges reproduce the results? (Scored on the three-tier evidence levels in §8.2; an L1 smoke test does not count as full reproduction.)
- **Scenario-driven demonstration and adaptability** (a strengthened requirement): **the submitted program must be able to actively design and execute challenge scenarios** to show the design works—for example, **dynamically adding classes** at run time, **simulating node disconnection / failure / reconnection**, injecting **distribution shift** or new scenarios—and demonstrate that the system still sustains collaboration and continual learning under these conditions. The demonstration must be triggerable by judges following the run instructions, or supported by an automation script / screen recording.

4.4 Axis D: Creativity (25%)

Judges will examine:

- Originality of the knowledge representation.
- Imagination in cross-domain integration.
- Future extensibility and research value.
- A distinctive interpretation of the concept of "collective intelligence".

4.5 Per-Axis Scoring Anchors (for judge calibration)

The table below offers **reference anchors**, not a rigid formula, to help different judges align their scales; judges may still deliberate on the evidence. Each axis runs 0–25, corresponding roughly to:

Axis	Low ($\approx 0-8$)	Medium ($\approx 9-17$)	High ($\approx 18-25$)
A Sustainability	No comparison baseline, or cites only TDP / peak spec without stating assumptions	A clear baseline (A or B) with a cost inventory, assumptions broadly reasonable, but some items missing or uncertainty undisclosed	Rigorous comparison; communication / compute / energy inventoried to the extent claimed (including on-device updates); transparent assumptions; uncertainty and counter-cases disclosed
B Integration & feasibility	Only vague talk of "integrable in future", with no path, no	An integration path and dependency account that maps to one	Concrete path; restrained, clearly versioned dependencies; honest

	dependency inventory, no barrier identification	industrial setting, but with over-optimistic barrier identification	about deployment barriers and interface-design awareness
C Technical maturity	Unstable system or no evidence of execution; no scenario demonstration; no fairness table for ΔG , or a clearly unfair comparison	Runs; has a baseline comparison, a ΔG fairness table, and a basic scenario demonstration, but insufficient ablation or reproducibility only at L1	Runs stably; can actively demonstrate multiple challenge scenarios (added classes / disconnection, etc.); complete ΔG fairness table; ablations present; reproducibility at L2 / L3
D Creativity	Existing methods applied directly, with no novelty	Some ingenuity in knowledge representation or scenario	An original knowledge-exchange mechanism, cross-domain imagination, or a distinctive interpretation of "collective intelligence"

The effect of violations or hard errors is handled separately under §5.4 (knowledge-exchange violations) and §6.1 (minimum evidence); these may deduct marks across axes or lead to disqualification, and are not bounded by the anchors above.

5. Knowledge-Exchange Rules

5.1 Prohibition on Transmitting Raw Data and on Data Tunnelling

The following must not be transmitted between nodes, directly or indirectly:

- Raw input data themselves—e.g. images, audio, text, sensor sequences, data fragments (whether cropped, sampled, or excerpted)—or their equivalent copies.
- Representations, features, compressed bundles, or serialised fragments from which the main content of the raw input could be reconstructed by reasonable means.
- Information that carries the semantics of raw data via metadata, prompts, file names, indices, external links, time, coordinate encodings, or any other side channel.
- Any data or signal that participates in a machine-readable exchange between nodes but is not disclosed in the Exchange Artefact Table.

This Challenge adopts a **principle of substantive review**: the judgement will not rest solely on how a participant names a data type.

5.2 Auditability of Knowledge Exchange

Participants must submit, in the research document, an **Exchange Artefact Table** covering all machine-readable exchanges between nodes. For each artefact, at minimum the following must be disclosed:

Field	Required	Description
artifact_name	✓	Human-readable name
purpose	✓	Purpose, e.g. distillation target / routing hint / model delta
direction	✓	Source → target node
frequency	✓	Exchange frequency or condition
size_per_message	✓	Size per message (bytes)
granularity	✓	per-sample / per-batch / per-class / aggregated to model level
contains_metadata	✓	Whether it contains time, coordinates, file names, prompts, IDs, external links
reconstruction_risk	✓	Self-assessed low / medium / high
risk_reason	✓	Why it is not raw data or a reconstructible representation (mandatory when low)
privacy_measure	△	Design measures such as compression, quantisation, noise addition, clipping, frequency limiting
notes	△	Additional remarks

A sample artefact does not mean automatic compliance. The distillation target / routing hint / model delta listed in the purpose field above are only examples of use, and are not uniformly lawful. The following artefacts are **assumed to be at least medium risk unless evidence to the contrary is provided**: sample-level embeddings, per-sample gradients or logits, reversibly compressed features, and summaries that contain raw prompts / text fragments or audio features. An artefact marked `reconstruction_risk = low` without a `risk_reason` filled in may not be treated as low-risk.

The organisers reserve the right to request supplementary materials, to spot-check, or to seek clarification on site.

5.3 Model-Initialisation Rules

- **Permitted:** all nodes using the same **public** pre-trained model or public dataset as a common starting point (the source must be disclosed in the research document, and the relevant licence complied with). This is precisely the lawful foundation that gives nodes a shared representational basis from which to learn from one another.

- **Prohibited:** using a model trained on the **union of multiple nodes' private data** as the initialisation weights—this amounts to sharing raw data by another name and renders the `local_only` baseline meaningless.
- **Pre-training purity:** the pre-trained model used as the common starting point must not already contain, or be specifically targeted at, domain data for this Challenge's target task (so that collaboration gain is not merely the reawakening of the model's existing memory rather than genuine edge self-learning). If the public data used are highly similar to a node's private data, this must be disclosed in the research document, together with its possible effect on the `local_only` baseline and on ΔG .
- This clause governs only the "model starting point before the Challenge begins"; **the collaborative and mutual learning carried out through knowledge exchange during the Challenge is not bound by this clause** (it follows §5.1 and §5.2).

5.4 Handling of Violations

- **Formal defects** (incomplete disclosure, missing fields): the team is asked to supplement the submission, and the defect is cleared once corrected.
- **Suspected side channel or reconstruction risk:** the corresponding Axis C marks are deducted, and supplementary explanation may be required. **An artefact marked high risk is not necessarily disqualified;** whether it constitutes a violation turns on whether the main raw content can reasonably be reconstructed and on whether disclosure was honest (concealment is treated more severely). Judges may first enter a questioning / spot-check process before ruling.
- **Clear violations** (directly transmitting raw input, equivalent copies, exchanges of reasonably reconstructible content, or deliberately concealing exchanged items): this results in disqualification.
- The organisers retain final discretion over violation determinations; a penalised team may submit one written appeal within three working days of notification.

6. Judging Principles

- The preliminary round is **judged primarily by human reviewers**, not by a fully automated benchmark.
- The more reproducible a submission, the more readily it earns a high score and priority for advancement.
- Creativity can compensate for some shortfall in integration feasibility and technical maturity, but **cannot substitute for the basic requirements of sustainability and privacy**.
- The organisers reserve the right to request supplements, to question, or to spot-check; teams that refuse to cooperate may have the corresponding axis marks reduced or be disqualified.
- Self-reported figures such as Collaboration Gain are **not compared across teams in absolute terms**; judges focus on the rigour of the comparison design (reasonableness of the data split, fairness of the baseline, completeness of ablations).

6.1 Minimum Evidence Requirements

Every submission must, at minimum, satisfy:

1. Being able to state the number of nodes, the roles of nodes, the timing of exchanges, and what is exchanged.
2. Providing reviewable execution evidence to **corroborate** that the system has actually run and that exchange between nodes did occur (for judges to spot-check, not a requirement for a full re-run): at minimum, a short **exchange summary** (listing, per round, the artefacts, direction, size, count, and corresponding code location), **plus at least one machine-generated piece of evidence** to avoid a purely hand-written summary—this may be a snippet of execution output, an automatically generated JSON / CSV exchange summary, checkpoint metadata, a communication-trace summary, or a screenshot / short recording (all lightweight, with no full re-run required). Full execution logs / communication traces / checkpoints are stronger evidence, verified at spot-check.
3. Providing a minimum comparison against a `local_only` baseline, or a reasonable explanation of why one cannot be provided (e.g. the scenario inherently cannot be split into a single-machine comparison, or real-hardware constraints).
4. Submitting a complete Exchange Artefact Table.
5. Submitting the measurement or estimate of at least one sustainability metric.

7. Format and Timeline (Single Track)

This Challenge is **single-track**; all teams compete under one common set of rules and scoring criteria.

Stage	Date	Content
Submission opens	July 2026	The online submission platform opens and begins accepting preliminary-round entries. You will be notified by email once the platform is open.
Submission deadline	2026-08-24 23:59 BST	The platform closes; judging begins immediately
Preliminary judging	2026-08-24 – mid-September	The judging panel scores all four axes (A/B/C/D, 25% each) in full, with spot-checks of the execution evidence, and ranks teams to select those advancing
Semi-finalists announced	Mid-September 2026	The 8 teams advancing to the semi-final are announced
Semi-final on-site presentation	2026-09-28	The 8 teams present in person at the University of Cambridge ; the judges select 3 teams to advance to the final
Final	2026-09-29	The 3 finalist teams give their final presentation and Q&A at the University of Cambridge; rankings are decided

- The preliminary round is entirely online, submitted via the platform (see §8, Deliverables).
- **The semi-final** is an on-site presentation: the 8 teams must travel to the University of Cambridge in person on 28 September to present and take questions; on the day, the judges select 3 teams to advance to the final.
- **The final** is held on 29 September at the University of Cambridge: the 3 teams give their final presentation and in-depth Q&A, and rankings are decided.
- **An offline-first approach is recommended for the on-site presentations:** the organisers guarantee only power and basic projection, and teams should bring whatever network equipment they need.

8. Deliverables

8.1 Preliminary-Round Submission Package (all three are mandatory)

#	Deliverable	Requirements
1	Research document	Core idea, knowledge-exchange design, learning mechanism, sustainability considerations; system architecture diagram; Exchange Artefact Table (per §5.2); the comparison for the chosen Axis A baseline (A / B), and the minimum local_only baseline comparison or reasonable alternative per §4.3 / §6.1 (including the Collaboration-Gain Fairness Disclosure Table, per Appendix A.1); sustainability measurement or estimate (at least one of energy / communication volume / compute cost); integration-feasibility narrative (industrial touchpoints and adoption path); disclosure of model initialisation and pre-existing IP (per §5.3); references / citations (the key references or papers relevant to the method must be listed; format is free; see Appendix D)
2	Presentation	A summary account of the solution for judges to understand quickly; qualifying teams will build on this for the on-site semi-final and final presentations (which may be updated beforehand)
3	Code	Executable source code, written mainly in Python (other high-level languages such as JavaScript are acceptable; performance / real-hardware components may use C / C++ / Rust, etc.); with run instructions (dependency versions, environment requirements, re-run steps) enabling judges to reproduce the results; must be able to actively run at least one scenario-driven demonstration (e.g. dynamically adding classes, simulating node disconnection / failure / reconnection, injecting distribution shift) to show the design still works under challenge conditions (triggerable by judges following the instructions, or supported by an automation script / screen recording); team information and the licence statement to be included

8.2 Code and Reproducibility

- **The main implementation should be primarily in Python** (other high-level languages such as JavaScript are acceptable); key components needed for performance or real-hardware deployment may use compiled languages such as **C / C++ / Rust**.
- Whatever the language, complete build and run instructions must be included, with an entry point judges can run through (e.g. a Python entry point, a build script, or a smoke test) to ensure reproducibility.
- The run instructions must cover: the required environment and dependencies (with versions), installation steps, how to start the multiple nodes, and how to reproduce the key results in the research document.
- Where hardware constraints prevent a full re-run (e.g. a specific edge device is needed), this must be clearly flagged in the run instructions, with reviewable execution evidence provided instead (logs / recording / screenshots). A team is not penalised merely for hardware constraints, but **a team that cannot provide any evidence at all that the system has actually run** will lose substantial marks.
- **Three-tier reproducibility evidence levels** (the Axis C reproducibility sub-criterion is scored on these tiers):
 - **L1:** a smoke test runs the multi-node flow through, but does not reproduce the figures in the research document.
 - **L2:** the key small-scale results can be re-run in a general environment.
 - **L3:** the main results can be re-run, or complete hardware-measurement evidence is provided.
 - L1 does not count as full reproduction; a hardware-constrained team that corroborates with reviewable evidence is not further penalised for being unable to reach L3.

8.3 Semi-Final and Final On-Site Deliverables

- **Semi-final (28 September, 8 teams):** present on site at Cambridge, covering the core of the solution, evidence of collaboration gain, sustainability and integration feasibility, and take questions from the judges; 3 teams are then selected to advance to the final.
- **Final (29 September, 3 teams):** give the final presentation and in-depth Q&A; rankings are decided.
- If the presentation and code have been updated since the preliminary round, an updated version may be submitted on site with the differences explained.

9. Academic and Industry Expectations

The judging panel particularly hopes to see work with the following qualities:

- **First-principles thinking:** a deep understanding of the nature of knowledge exchange, especially the design trade-offs among communication, compute, and energy.
- **Methodological portability:** the solution should generalise theoretically across scenarios and modalities.

- **Rigorous experimental controls:** a clear ablation or controlled account of the source of collaboration gain.
- **Continual-learning capability:** the ability to face new classes, new scenarios, and long-running conditions.
- **Emergent collaborative value:** the system's overall performance after collaboration should be explained in terms of why it surpasses a single node or a naive concatenation.
- **A vision for sustainable deployment:** the solution should map to concrete industrial settings for energy saving, waste reduction, or resource savings.

10. Post-Challenge Openness and Follow-Up Collaboration

- Outstanding work may be invited to take part in industry exchange, forum showcases, and follow-up collaboration discussions (deep R&D, industrial integration, real-hardware validation, etc.).
- This Challenge does not guarantee publication or CISL academic endorsement; however, opportunities for publication or white-paper collaboration may be explored separately, where there is clear mutual value and alignment.
- Intellectual property is handled under **Appendix F: IP Policy**.

11. Eligibility and Team Rules

- **Eligibility:** Any researchers, postdoctoral researchers from a UK university or current students enrolled at a UK university may participate, with no restriction on institution or programme (undergraduate, master's, or doctoral). Each team must have a lead representative who is based in or able to travel to Cambridge from elsewhere within the UK (reasonable travel expenses will be reimbursed). All team members must be 18 years of age or above.
- **Submission deadline:** 2026-08-24 23:59 (BST, British Summer Time).
- **Team size / multiple teams per person / entries per team:** There are no restrictions on team size or composition, provided at least one team member is a researcher, postdoctoral candidate or current student at a UK university at the time of submission. Those who wish to enter multiple solutions should submit separate entries for each concept.

12. Appendices

Appendix A: Suggested Collaboration-Gain Metrics

The organisers suggest participants compute the following metrics, which may serve as support for Axis A / C:

- **Collaboration Gain (ΔG_{local}):** the improvement of the collaborative version over the `local_only` baseline.

- Collaboration Gain (ΔG_{best}): the improvement of the collaborative version over the best single-node result.
- adaptation_gain: the adaptation gain when facing new classes, distribution shift, or node changes.
- retention_gain: the ability to retain prior knowledge.
- communication_efficiency: the efficiency of gain relative to communication cost.
- gain_per_joule: the efficiency of collaboration gain relative to energy cost (an important support for Axis A).

The formula and method of computation may be chosen by participants and need only be explained in the research document. The organisers do **not** automatically determine ranking or compliance from any single metric.

Appendix A.1: Collaboration-Gain Fairness Disclosure Table (mandatory for ΔG)

To prevent "strawman gain" (deliberately weakening local-only to inflate collaboration gain), any team reporting ΔG must include the table below in the research document. The organisers do **not** designate an official dataset or a unified formula, but require the **disclosure** of the following items so that judges can determine whether the comparison is fair:

Field	Requirement
task_metric	The main score, e.g. accuracy / F1 / mAP / MAE
node_count	The number of nodes and the role of each
data_split	The train / validation / test split method, random seed, and the sample count per node
non_iid_description	How class / scenario / time / domain skew is produced, and its strength
test_set_scope	local test, global held-out test, or both; test samples must not enter any training set
local_only_budget	The model, epochs / steps, data volume, and tuning effort for local-only
collective_budget	The model, epochs / steps, exchange rounds, and tuning effort for collective
per_node_scores	The local-only and collective scores for each node
macro_avg_and_worst_node	The macro average and the worst-node score (not just the best or the overall average)
delta_g_formula	The ΔG formula actually used, and the limits of comparability

If `local_only_budget` and `collective_budget` are not aligned, a `fairness_exception_reason` must be filled in; without it, ΔG may not be treated as high-confidence evidence (see §4.3). A team unable to run local-only must explain why it does not apply, and indicate whether its alternative baseline is more lenient or more stringent than local-only.

Appendix B: Advanced Challenges (not scored; exploration encouraged)

1. Asymmetric knowledge distillation between heterogeneous nodes.
2. Byzantine-fault-tolerant collective intelligence.
3. Dynamic topology optimisation under communication constraints.
4. Mitigating catastrophic forgetting in a decentralised setting.
5. Collaborative learning on ultra-low-power (< 1 W) edge hardware.

Appendix C: Suggested Simulation Paths (not scored; for guidance only)

- Path 1: multiple containers simulating multiple edge nodes.
- Path 2: several laptops forming a local network.
- Path 3: a phone app or real-hardware devices.
- Path 4: low-power development boards (Jetson, Coral, Raspberry Pi, etc.).

Whichever path you take, preparing a version that can run offline is recommended.

Appendix D: Suggested Academic References

1. McMahan et al., *Communication-Efficient Learning of Deep Networks from Decentralized Data*.
2. Hinton et al., *Distilling the Knowledge in a Neural Network*.
3. Bilen et al., *Towards Universal Object Detection by Domain Attention*.
4. Kirkpatrick et al., *Overcoming Catastrophic Forgetting in Neural Networks*.
5. Snell et al., *Prototypical Networks for Few-shot Learning*.
6. Lian et al., *Can Decentralized Algorithms Outperform Centralized Algorithms?*.
7. Kairouz et al., *Advances and Open Problems in Federated Learning*.

Note: the titles in this list retain the original (American) spelling of their publications.

Appendix E: Sustainability-Measurement Guidance (recommended practice)

To help participants present Axis A concretely, this appendix provides measurement / estimation methods and **the correct way to estimate energy**. Those without an energy meter are not

penalised; an estimate is treated on a par with measurement as long as its logic is rigorous—but **providing no sustainability measurement or estimate at all** loses marks on Axis A.

E.1 Overview of Measurement / Estimation Methods (choose any)

Category	Method	Applicable when
Measured energy	Using a wattmeter / NVIDIA nvidia-smi / Intel RAPL / Arm Energy Probe, etc.	Hardware support is available
Estimated energy	See the total-energy model in E.2 (compute + memory access + communication)	No instrument available
Communication cost	Cumulative exchanged bytes, message frequency, exchange rounds	All settings
Compute cost	Model parameter count, FLOPs / MACs, inference latency, training time	All settings
Environmental-benefit narrative	If deployed in the target setting, the estimated annual energy saving / waste reduction	Application-leaning work

E.2 The Correct Way to Estimate Energy (important)

The minimum submission requirement still follows §6.1 / §8.1: providing at least one sustainability measurement or estimate, with a comparison baseline, is compliant. The three energy terms below **need a full treatment only when a team claims "end-to-end energy saving", uses gain_per_joule, or relies on energy figures as primary Axis A evidence**; a missing term must be flagged as a limitation—failing to estimate all three fully **does not constitute non-compliance**, but does affect the rigour of the Axis A estimate.

Energy estimation is easy to get wrong. **The complete end-to-end energy = compute energy (including inference and on-device learning / updates) + memory-access energy + communication energy.** Note in particular: this Challenge centres on **self-learning**, and the energy of on-device training / updates (backpropagation, frequent fine-tuning) is often **greater than that of inference** and is precisely the cost collaboration adds, so **inference alone must not be counted**. The three pitfalls below are what judges use to examine the rigour of an estimate:

(1) Do not take TDP directly as actual power. TDP (Thermal Design Power) is a thermal-design ceiling, **not** a device's actual power during a given inference. $\text{TDP} \times \text{inference time}$ can serve only as a **loose upper bound (a rough estimate)**, not as actual energy. A more reasonable approach:

- Use the device's **actual / typical active power** under that workload, or multiply TDP by a justified **utilisation factor** (and state its source).
- State explicitly the voltage / frequency / quantisation precision assumed in the estimate (the power difference between INT8 and FP16 is large).

(2) Do not use FLOPs alone; memory-access energy (the Memory Wall) is usually the larger share. Relying solely on $\text{FLOPs} \times \text{J/MAC}$ severely underestimates energy consumption, because the energy of one DRAM access can be **about two orders of magnitude** (over a hundredfold) that of

one MAC operation. On memory-bound edge models, moving data is often more energy-consuming than arithmetic. A more reasonable approach:

- Include **memory-access volume** (the read/write bytes of weights / activations) in the estimate, with a corresponding **J/byte coefficient** (literature may be cited, e.g. Horowitz's energy figures as a baseline).
- At minimum, **acknowledge and discuss** the impact of memory-access energy in the document, even if only a rough coefficient is given.

(3) Communication energy must be counted—it is the cost unique to the collaborative approach.

The energy efficiency of wireless transmission on edge devices (WiFi / LTE / BLE) is typically **1–2 orders of magnitude** lower than local compute: the energy to send one bit may far exceed that of one MAC. If a solution **exchanges frequently** (model deltas / embeddings), its communication energy may actually **exceed** the compute energy it saves, making "collaboration" a net loss in sustainability. A more reasonable approach:

- Communication energy $\approx$ total exchanged bytes $\times$ the J/bit (or J/byte) coefficient of that wireless interface, multiplied by the number of exchange **rounds / frequency**.
- Show explicitly, in the Axis A comparison, whether the compute energy collaboration saves is negated by the additional communication energy incurred. This is exactly the core question Baseline B (gain_per_joule) is meant to answer.

E.3 Estimation-Credibility Grading (judges assess rigour by this)

Credibility	Condition
Low	Cites only TDP, TOPS/W, or peak vendor specs, without stating assumptions
Medium	For the sustainability metric claimed, states the model's FLOPs, memory-access volume, inference or on-device learning / update time , quantisation precision, estimation assumptions, and uncertainty; if end-to-end energy is claimed, communication energy is included
High	Measures power on a real / simulated device or via system tools, disclosing the measurement conditions and the voltage / frequency settings

Judges do not require "High" credibility; but at the same figures, an estimate with **transparent assumptions, all three energy terms present, and clearly stated credibility** earns a higher Axis A rating.

E.4 Advanced Discussion (bonus; not required)

The following are **encouraged bonuses**, not hard thresholds (most teams cannot reliably measure static power):

- **Static / idle energy and Time-to-Accuracy:** if a solution uses a synchronous protocol, multiple communication rounds, or a persistent connection, discussing the static energy accumulated while "converging to the target accuracy"—through prolonged radio / CPU active time spent waiting for synchronisation—is recommended; this may offset or even exceed the compute

energy saved. Assessing collaboration efficiency by the "total energy to reach a usable state", rather than the energy of a single inference alone, is closer to physical reality.

Appendix F: Intellectual Property Policy

This Challenge's IP policy adopts the following five principles:

1. **Participants own by default:** all code, designs, and ideas created during the Challenge are owned by the participants; the Challenge claims no compulsory assignment.
2. **A licence granted to the organisers:** participants grant the organisers (MICROIP, Arculus System, and the University of Cambridge Institute for Sustainability Leadership) a **non-exclusive, royalty-free** licence, solely for the display purposes of marketing, event reporting, and the promotion of future activities, not constituting a transfer of ownership.
3. **Disclosure of pre-existing IP:** if participants use pre-existing code, libraries, models, or datasets, they should disclose this in the research document and ensure compliance with the relevant open-source licence.
4. **Commercialisation negotiated separately:** if the organisers, judges, or partners wish to commercialise any solution, incorporate it into a product, or offer employment, a **separate agreement** must be signed with the team after the Challenge.
5. **Confidentiality:** the Challenge does not require participants to disclose any trade secret; if the organisers provide any proprietary data or resources, a separate NDA will be signed.

Appendix G: External Communication and Composition of the Judging Panel

- **Composition of the panel:** in principle, a joint panel of representatives from the University of Cambridge Institute for Sustainability Leadership and industry; the final composition will be confirmed before the official announcement.
- **Judging feedback:** all advancing teams may obtain brief written feedback after the Challenge.
- **Follow-up collaboration:** teams that perform outstandingly may be invited into extended collaboration (deep R&D, industrial integration, etc.), with details negotiated case by case.

13. Conclusion

What the organisers hope to see is not who is most adept at assembling a technical stack, but who can genuinely answer this question:

Under the constraints that raw data cannot be exchanged and that energy and compute resources are limited, how can multiple self-learning edge nodes, through knowledge exchange, form a collective intelligence that delivers measurable collaboration gain, maps to industrial-integration settings, and demonstrates sustainability value?

14. Enquiries

Should you have any questions about the technical details of this Challenge that have not been answered above, kindly navigate to the online submission platform (<https://challenge.arculussystem.com>) for further support. Navigate to the links available in the footer to view frequently asked questions or to submit a question of your own.



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